

WUSHAOLING, China

WMO: 527870

Lat: 37.1988N

Lon: 102.8680E

Elev: 3044

StdP: 69.72

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.6	-18.8	-30.7	0.3	-8.4	-28.7	0.4	-8.3	14.2	-13.9	13.1	-12.6	5.3	320	0.867

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.4	20.2	11.2	18.7	10.5	17.2	9.7	13.6	17.4	12.5	16.2	11.5	15.0	5.9	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.3	13.0	14.8	11.2	12.1	13.8	10.2	11.3	13.2	49.4	17.5	45.9	16.4	42.7	14.9	17.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.4	11.9	10.6	DB	-23.3	23.3	2.5	1.5	-25.0	24.3	-26.5	25.2	-27.9	26.1	-29.6	27.1	(4)
(5)			WB	-23.6	15.2	2.4	1.1	-25.3	16.0	-26.7	16.6	-28.0	17.3	-29.7	18.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	1.3	-10.7	-8.5	-4.0	1.7	6.1	10.3	12.6	11.5	7.4	1.9	-3.8	-8.8		
	DBStd	8.85	4.35	4.50	4.76	4.61	3.72	2.70	2.62	2.51	2.98	3.66	4.13	4.04		
	HDD10.0	3357	641	518	434	251	129	28	6	13	87	253	415	583		
	HDD18.3	6201	899	752	692	499	381	242	180	211	328	511	665	841		
	CDD10.0	199	0	0	0	2	7	36	85	60	9	1	0	0		
	CDD18.3	1	0	0	0	0	0	0	1	0	0	0	0	0		
	CDH23.3	3	0	0	0	0	0	0	3	0	0	0	0	0		
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0		
(14) Wind	WSAvg	5.3	5.0	5.4	6.0	6.1	5.8	5.2	5.0	5.0	5.1	5.0	4.9	4.9		
(15) Precipitation	PrecAvg	390	1	3	9	18	38	64	89	87	58	18	4	1		
	PrecMax	555	3	12	34	46	89	131	172	178	164	34	11	4		
	PrecMin	231	0	0	1	5	9	26	42	32	17	4	0	0		
	PrecStd	84	1	3	7	9	22	26	33	35	27	9	3	1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.0	6.7	11.5	16.9	18.3	20.0	23.1	21.6	18.3	14.3	8.6	3.9		
		MCWB	-4.1	-2.4	0.5	4.2	6.5	8.7	12.8	12.7	9.4	5.2	-0.8	-3.9		
	2%	DB	0.7	3.2	7.6	13.4	15.8	18.2	21.0	19.4	15.6	11.4	6.0	1.1		
		MCWB	-5.9	-3.8	-1.0	3.0	5.6	8.7	11.9	11.9	7.9	3.4	-1.7	-5.3		
	5%	DB	-1.8	0.7	5.1	10.9	14.2	16.7	19.2	17.8	13.8	9.3	3.9	-0.8		
		MCWB	-7.3	-5.1	-2.1	2.0	4.9	8.1	11.3	11.0	7.6	2.5	-2.7	-6.5		
	10%	DB	-4.0	-1.6	2.7	8.7	12.5	15.3	17.6	16.2	12.2	7.2	2.1	-2.6		
		MCWB	-8.6	-6.5	-3.2	1.0	4.2	7.4	10.7	10.4	6.8	1.6	-3.6	-7.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-4.0	-1.8	1.3	5.6	8.1	12.4	15.2	15.4	12.8	6.5	0.1	-3.8		
		MCDB	3.7	5.2	10.1	12.7	14.7	16.0	20.1	18.2	15.2	11.5	6.5	3.3		
	2%	WB	-5.8	-3.6	-0.6	4.1	6.8	10.7	13.8	13.6	10.3	4.8	-1.2	-5.2		
		MCDB	0.5	2.9	6.4	11.4	13.4	14.4	18.5	16.7	12.9	9.2	4.8	0.8		
	5%	WB	-7.3	-5.0	-1.8	2.8	5.8	9.6	12.7	12.6	8.9	3.5	-2.3	-6.3		
		MCDB	-2.1	0.4	4.1	9.3	12.0	13.7	17.0	15.8	11.8	7.5	3.1	-1.1		
	10%	WB	-8.6	-6.4	-2.8	1.6	5.0	8.7	11.6	11.6	7.8	2.4	-3.3	-7.4		
		MCDB	-4.2	-1.9	2.1	7.6	11.0	13.2	15.7	14.6	10.6	6.4	1.3	-2.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.1	9.4	8.7	8.6	8.5	7.6	7.4	7.3	7.3	7.9	8.7	9.0		
		MCDBR	11.4	12.3	11.5	11.4	11.2	9.8	10.1	9.8	9.5	10.2	10.6	10.7		
	5% WB	MCWBR	7.3	7.6	6.3	5.8	5.5	4.7	4.8	4.9	5.1	5.3	5.7	6.7		
		MCWBR	11.1	11.8	10.5	10.3	9.7	7.8	8.6	8.0	7.9	9.1	9.9	10.5		
(40) Clear-Sky Solar Irradiance	taub		0.267	0.319	0.383	0.426	0.427	0.422	0.406	0.414	0.387	0.345	0.285	0.257		
	taud		2.343	2.161	1.985	1.912	1.989	2.068	2.188	2.142	2.176	2.212	2.334	2.390		
	Ebn at Noon		928	904	871	853	858	860	870	855	859	863	895	919		
	Edh at Noon		91	125	166	189	178	165	145	148	135	118	91	81		
(44) All-Sky Solar Radiation	RadAvg		2.94	3.77	4.80	5.60	5.99	5.91	5.69	5.19	4.22	3.88	3.19	2.75		
	RadStd		0.20	0.18	0.29	0.31	0.33	0.34	0.50	0.36	0.32	0.31	0.17	0.11		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47) Regional (5 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+81	+32		

Nomenclature: See separate page